

```

create table S_Customer(
Customer_ID varchar(50) primary key,
Customer_Name varchar(50),
Segment varchar(50)

);
INSERT INTO S_Customer(Customer_ID, Customer_Name, Segment)
select distinct
Customer_ID,
Customer_Name,
Segment
from Central_Superstore
Where Customer_ID is not null;

```

```

create table S_Product(
Product_ID varchar(50) primary key,
Product_Name varchar(255),
Category varchar(50),
Sub_Category varchar(50)
);
INSERT INTO S_Product(Product_ID, Product_Name, Category, Sub_Category)
select
Product_ID, max(Product_Name), max(Category), max([Sub_Category])
from Central_Superstore
Where Product_ID is not null
group by Product_ID;

```

```

create table S_Location(
Postal_Code varchar(100) primary key,
City varchar(100),
State varchar(100),
Country varchar(50),
Region varchar(50)
);
INSERT INTO S_Location(Postal_Code, City, State, Country, Region)
Select
CAST(Postal_Code as varchar(20)), max(City), max(State), max(Country), max(Region)
from Central_Superstore
Where Postal_Code is not null
group by Postal_Code;

```

```

create table S_ShipMode(
Ship_Mode_ID int identity(1,1) Primary key,
Ship_Mode varchar (50)
);
INSERT INTO S_ShipMode(Ship_Mode)
select Distinct Ship_Mode
from Central_Superstore
Where Ship_Mode is not null;
create table Fact_Orders(
Row_ID int primary key,
Order_ID varchar(50) not null,
Order_Date Date,
Ship_Date Date,
Customer_ID varchar(50),

```

```

Product_ID varchar(50),
Postal_Code varchar(100),
Ship_Mode_ID int,
Sales float,
Quantity int,
Discount float,
Profit float,
CONSTRAINT FK_Orders_Customer FOREIGN KEY (Customer_ID) REFERENCES
S_Customer(Customer_ID),
CONSTRAINT FK_Orders_Product FOREIGN KEY (Product_ID) REFERENCES S_Product(Product_ID),
CONSTRAINT FK_Orders_Location FOREIGN KEY (Postal_Code) REFERENCES S_Location(Postal_Code),
CONSTRAINT FK_Orders_ShipMode FOREIGN KEY (Ship_Mode_ID) REFERENCES
S_ShipMode(Ship_Mode_ID)
);
INSERT INTO Fact_Orders (Row_ID, Order_ID, Order_Date, Ship_Date, Customer_ID, Product_ID,
Postal_Code, Ship_Mode_ID, Sales, Quantity, Discount, Profit)
SELECT
r.Row_ID,
r.Order_ID,
DATEADD(day, r.Order_Date, '1899-12-30'),
DATEADD(day, r.Ship_Date, '1899-12-30'),
r.Customer_ID,
r.Product_ID,
r.Postal_Code,
sm.Ship_Mode_ID,
r.Sales,
r.Quantity,
r.Discount,
r.Profit
FROM Central_Superstore as r
INNER JOIN S_ShipMode as sm
ON r.Ship_Mode = sm.Ship_Mode;
select count(distinct Order_ID) as Total_Orders,
count(Row_ID) as Total_Items_Sold,
sum(Sales) as Total_sales,
sum (Profit) as Total_Profit
from Fact_Orders;

select
sum(f.Sales) as Total_sales,
sum (f.Profit) as Total_Profit,
sum(f.Quantity) as Total_quantity
from Fact_Orders as f
inner join S_Product as p
on p.Product_ID= f.Product_ID
group by Category
order by Total_Profit desc;
select top 5
c.Customer_Name,
c.Segment,
sum(f.Sales) as Total_Spent
from Fact_Orders f
JOIN S_Customer c on f.Customer_ID = c.Customer_ID
group by c.Customer_Name, c.Segment
order by Total_Spent desc;
select
l.State,
sum(f.Sales) as Total_Sales,

```

```

    sum(f.Profit) as Total_Profit
from Fact_Orders as f
INNER JOIN S_Location as l
    ON f.Postal_Code = l.Postal_Code
group by l.State
order by Total_Profit desc;

```

```

select
    l.State,
    sum(f.Sales) as Total_Sales,
    sum(f.Profit) as Total_Profit
from Fact_Orders as f
INNER JOIN S_Location as l
    ON f.Postal_Code = l.Postal_Code
group by l.State
order by Total_Profit asc;
select

```

```

    sm.Ship_Mode,
    count(f.Order_ID) as Total_Orders,
    sum(f.Sales) as Total_Sales,
    sum(f.Profit) as Total_Profit
from Fact_Orders as f
inner join S_ShipMode as sm
    on f.Ship_Mode_ID = sm.Ship_Mode_ID
group by sm.Ship_Mode
order by Total_Sales desc;

```

```

select
    l.State,
    sum(f.Sales) as Total_Sales,
    sum(f.Profit) as Total_Profit,
    avg(f.Discount) * 100 as Avg_Discount_Percentage
from Fact_Orders as f
INNER JOIN S_Location as l
    on f.Postal_Code = l.Postal_Code
group by l.State
order by Total_Profit asc;
select

```

```

    p.Sub_Category,
    sum(f.Sales) as Total_Sales,
    sum(f.Profit) as Total_Profit,
    avg(f.Discount) * 100 as Avg_Discount_Percentage
from Fact_Orders as f
INNER JOIN S_Product as p
    on f.Product_ID = p.Product_ID
group by p.Sub_Category
order by Total_Profit asc;
select

```

```

    year(f.Order_Date) as Order_Year,
    count(DISTINCT f.Order_ID) as Total_Orders,
    sum(f.Sales) as Total_Sales,
    sum(f.Profit) as Total_Profit
from Fact_Orders as f
group by YEAR(f.Order_Date)
order by Order_Year asc;
create view View_State_Performance as
select

```

```

    l.State,

```

```

sum(f.Sales) as Total_Sales,
sum(f.Profit) as Total_Profit,
avg(f.Discount) * 100 as Avg_Discount_Percentage
from Fact_Orders as f
INNER JOIN S_Location as l
on f.Postal_Code = l.Postal_Code
group by l.State;
select * from View_State_Performance order by Total_Profit desc;

```

```

create procedure sp_Get_State___Report
@StateName varchar(50)
as
begin
select
l.State,
count(DISTINCT f.Order_ID) as Total_Orders,
sum(f.Sales) as Total_Sales,
sum(f.Profit) as Total_Profit,
avg(f.Discount) * 100 as Avg_Discount_Percentage
from Fact_Orders as f
INNER JOIN S_Location as l
ON f.Postal_Code = l.Postal_Code
WHERE l.State = @StateName
GROUP BY l.State;
END;
EXEC sp_Get_State___Report @StateName = 'Michigan';

```

```

WITH Customer_Sales_CTE AS (
SELECT
Customer_ID,
SUM(Sales) AS Total_Spent
FROM Fact_Orders
GROUP BY Customer_ID
)
SELECT TOP 5 *
FROM Customer_Sales_CTE
ORDER BY Total_Spent DESC;

```

```

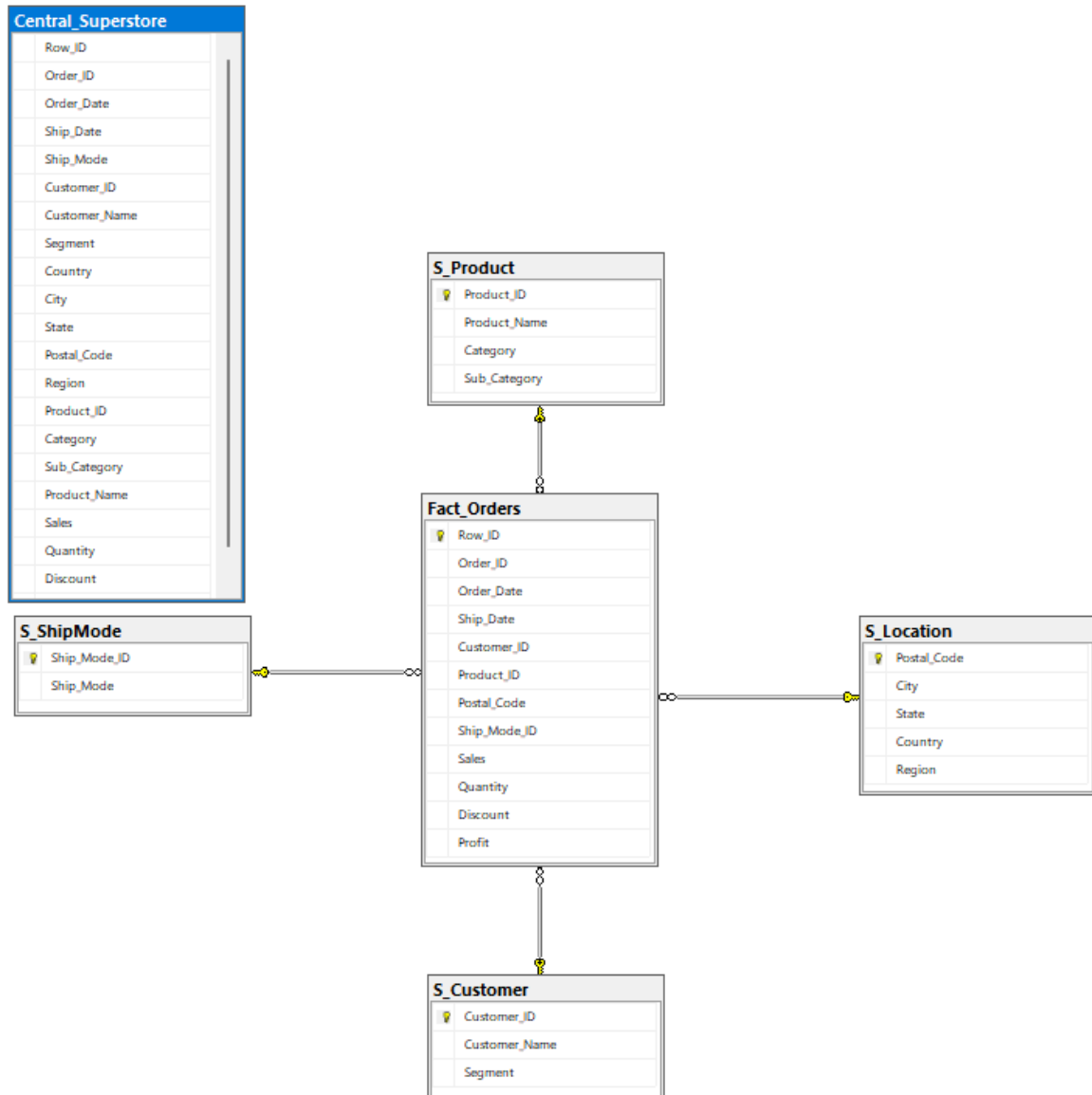
WITH Profit_Status_CTE AS (
SELECT
Order_ID,
Profit,
CASE
WHEN Profit > 0 THEN 'Profitable'
ELSE 'Loss'
END AS Status
FROM Fact_Orders
)
SELECT
Status,
COUNT(Order_ID) AS Total_Orders
FROM Profit_Status_CTE
GROUP BY Status;

```

```

SELECT
    Order_ID,
    Sales
FROM Fact_Orders
WHERE Sales > (SELECT AVG(Sales) FROM Fact_Orders);

```



Name: Adham Ahmed Mohamed Mohamed

Code: 21177183

Group: CAI5_DAT3_S6

#####